

Question	Answer	Marks
2(a)	total kinetic energy associated with random motion of molecules	M1
	plus total potential energy (of molecules) but potential energy is zero	A1
2(b)(i)	$W = p\Delta V$	C1
	$= 1.01 \times 10^5 \times 5.20 \times 10^{-5}$ $= (+)5.25 \text{ J}$	A1
2(b)(ii)	$V \propto T$ or $V/T = \text{constant}$	C1
	$1.24 / (273 + 20) = (1.24 + 0.520) / T$ $T = 416 \text{ K}$	A1
2(b)(iii)	$c = Q / m\Delta T$	C1
	$= 960 / (0.016 \times (416 - 293))$ $= 490 \text{ J kg}^{-1} \text{ K}^{-1}$	A1
2(c)	no change in volume so no work is done (by the gas)	B1
	(same temperature change so) same change in internal energy	B1
	less thermal energy needs to be supplied so c is less	B1

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